

# Explanation of Nature (2008) and Nelson (2007) Model

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## 1 Introduction

In this document, we will explain the equation structure of the size-structured consumer-resource model presented in Nature (2008) and Nelson (2007). We will use the equation formatting from the Nature supplement, which compartmentalizes certain rates (ingestion and death). Note that the equations we presented match those in Box 1 of the official 2008 Nature paper, with some minor expansions.

This model describes an ecosystem consisting of an algal resource abundance  $F(t)$  and a Daphnia consumer population, split into adult Daphnia  $A(t)$  and juvenile Daphnia  $J(t)$ . The algae have units of  $mgCL^{-1}$  while the juvenile and adult Daphnia have units  $\frac{\#individuals}{L}$ . The algae grow logistically and are consumed by juvenile and adult Daphnia according to a type II functional response. Nelson 2007 asserts that adult Daphnia consume more than juvenile Daphnia. Only adults reproduce, and within each stage, all individuals have the same feeding, developmental, and mortality rates. Our model assumes that a fixed proportion of consumed carbon is available for use, and that a fixed proportion of available carbon is used for reproduction. Juvenile maturation at time  $t$  depends on the amount of carbon ingested at time  $t - \tau(t)$  and a through-stage survival probability  $S(t)$ . Mortality rates for both juveniles and adults are assumed to be inversely proportional to their respective ingestion rates.

## 2 Parameters and Helper Functions

Before we begin, let us present our parameters in Table 1 and a list of helper functions in Table 2

Table 1: Parameters for the stage-structured consumer–resource model.

| Parameter  | Definition                                              | Units                           |
|------------|---------------------------------------------------------|---------------------------------|
| $q$        | Maximum per-capita rate of resource growth              | $\text{d}^{-1}$                 |
| $k$        | Resource carrying capacity                              | C                               |
| $I_J$      | Maximum juvenile ingestion rate                         | $\text{C N}^{-1} \text{d}^{-1}$ |
| $I_A$      | Maximum adult ingestion rate                            | $\text{C N}^{-1} \text{d}^{-1}$ |
| $f_h$      | Half-saturation constant in the functional response     | C                               |
| $\chi$     | Proportion of utilized carbon allocated to reproduction | dimensionless                   |
| $\gamma$   | Carbon required to produce one new offspring            | $\text{C N}^{-1}$               |
| $\sigma_J$ | Proportion of ingested carbon utilized by juveniles     | dimensionless                   |
| $\sigma_A$ | Proportion of ingested carbon utilized by adults        | dimensionless                   |
| $\mu_J$    | Juvenile mortality scalar                               | $\text{C N}^{-1} \text{d}^{-2}$ |
| $\mu_A$    | Adult mortality scalar                                  | $\text{C N}^{-1} \text{d}^{-2}$ |
| $w$        | Mass gain required to complete juvenile development     | $\text{C N}^{-1}$               |

Here, N denotes # individuals per liter, C denotes  $\text{mgCL}^{-1}$ , and d denotes days.

Table 2: Functions used in the stage-structured consumer–resource model.

| Functions                                        |                           |
|--------------------------------------------------|---------------------------|
| $g(t) = qF(t) \left(1 - \frac{F(t)}{k}\right)$   | Resource growth function  |
| $h(t) = \sigma_J \phi_J(t)$                      | Juvenile development rate |
| $\delta_J(t) = \frac{\mu_J}{\sigma_J \phi_J(t)}$ | Juvenile mortality rate   |
| $\delta_A(t) = \frac{\mu_A}{\sigma_A \phi_A(t)}$ | Adult mortality rate      |
| $\phi_J(t) = I_J \frac{F(t)}{F(t) + f_h}$        | Juvenile ingestion rate   |
| $\phi_A(t) = I_A \frac{F(t)}{F(t) + f_h}$        | Adult ingestion rate      |

### 3 Model Breakdown

In this section we shall go through each portion of our model, stating with the balance equations (1), then the stage duration(2), the birth rate (3), and lastly the maturation rate in (4) and 3.5.

### 3.1 Balance Equations

$$\frac{dF(t)}{dt} = \underbrace{g(t)}_{\text{logistic algae growth}} - \underbrace{\phi_J(t)J(t)}_{\text{ingestion by juveniles}} - \underbrace{\phi_A(t)A(t)}_{\text{ingestion by adults}} \quad (1a)$$

$$\frac{dJ(t)}{dt} = \underbrace{R_J(t)}_{\text{birth/recruitment}} - \underbrace{R_A(t)}_{\text{maturation}} - \underbrace{\delta_J(t)J(t)}_{\text{juvenile mortality}} \quad (1b)$$

$$\frac{dA(t)}{dt} = \underbrace{R_A(t)}_{\text{maturation}} - \underbrace{\delta_A(t)A(t)}_{\text{adult mortality}} \quad (1c)$$

In (1a), we have that the change in algae equals the growth rate minus the rate at which algae is ingested by juveniles and adults. In (1b), we have that the change in the juvenile population equals the rate of new births/recruitment minus the rate of juveniles maturing into adults and the rate of juvenile deaths. Lastly, in (1c), the change in adult population is the rate of newly matured juveniles minus the rate of adult deaths.

### 3.2 Stage Duration

We define the stage duration  $\tau(t)$  in terms of the following integral

$$w = \int_{t-\tau(t)}^t h(\xi) d\xi \quad (2)$$

where the integrand

$$h(\xi) = \sigma_J \phi_J(\xi)$$

is the rate of usable ingested carbon at time  $\xi$ . Thus the integral sums the total ingested carbon over the past  $\tau(t)$  days. And since  $w$  is the mass gain required to complete juvenile development,  $\tau(t)$  is defined as the duration spent in the juvenile stage by the group of individuals who mature at time  $t$ .

### 3.3 Birth Rate

Now, let us list then explain the birth rate equation.

$$R_J(t) = \frac{\chi}{\gamma} \sigma_A \phi_A(t) A(t) \quad (3a)$$

In (3a), let us first consider the proportion terms  $\chi$ ,  $\gamma$ , and  $\sigma_A$ .  $\chi$  is the proportion of utilized carbon allocated to reproduction while  $\gamma$  is carbon required to produce one new offspring.  $\sigma_A$  is the constant proportion of all ingested carbon that is able to be utilized by adults. Put together, the term

$$\frac{\chi}{\gamma} \sigma_A$$

scales the amount of carbon ingested by adults into a number of newly produced juveniles. It has units of  $[\frac{\# \text{individuals}}{mg C}]$ .  $\phi_A(t)$  is simply the rate of ingestion by an adult at time  $t$ , and thus  $\phi_A(t)A(t)$  is the rate of total carbon ingested at time  $t$ . After accounting for the proportion of carbon that can be utilized, the proportion allocated to reproduction, and the amount of carbon required per offspring,  $R_J(t)$  gives the total rate at which new juveniles enter the population.

### 3.4 Maturation Equation

Now, let us define the maturation equation  $R_A(t)$ , which is by far the most complicated term, as well as the survival probability equation used within it.

$$R_A(t) = R_J(t - \tau(t)) \frac{h(t)}{h(t - \tau(t))} S(t) \quad (4a)$$

$$= \frac{\chi}{\gamma} \sigma_A \phi_A(t) A(t - \tau(t)) S(t) \quad (4b)$$

$$S(t) = \exp \left\{ - \int_{t-\tau(t)}^t \delta_J(\xi) d\xi \right\}. \quad (4c)$$

### 3.5 Simplification/Expansion of maturation equation

Let us expand and rearrange (4a) to yield (4b). To do this, we shall expand  $h(t), h(t - \tau(t), R_J(t - \tau(t)))$  which yields

$$R_A(t) = \frac{\chi}{\gamma} \sigma_A \phi_A(t - \tau(t)) A(t - \tau(t)) \frac{\sigma_J \phi_J(t)}{\sigma_J \phi(t - \tau(t))} S(t)$$

Notice that the type II saturation functions have similar terms and cancel as follows

$$\phi_A(t - \tau(t)) \frac{\sigma_J \phi_J(t)}{\sigma_J \phi(t - \tau(t))} = I_A \frac{F(t - \tau(t))}{F(t - \tau(t)) + f_h} \frac{I_J \frac{F(t)}{F(t) + f_h}}{I_J \frac{F(t - \tau(t))}{F(t - \tau(t)) + f_h}} \quad (5)$$

$$= I_A \frac{F(t)}{F(t) + f_h} \quad (6)$$

Which is equal to  $\phi_A(t)$ . Thus  $R_A(t)$  can be rewritten as

$$R_A(t) = \frac{\chi}{\gamma} \sigma_A \phi_A(t) A(t - \tau(t)) S(t)$$

### 3.6 Explanation of Maturation Equation

In this section, I will paraphrase the argument used by W. S. C. Gurney and R. M. Nisbet in the “The Systematic Formulation of Delay-Differential Models of Age or Size Structured Populations” from 1983. Their explanation uses a conveyor-belt analogy to represent the maturation of juveniles into adults, as seen in Figure 1.

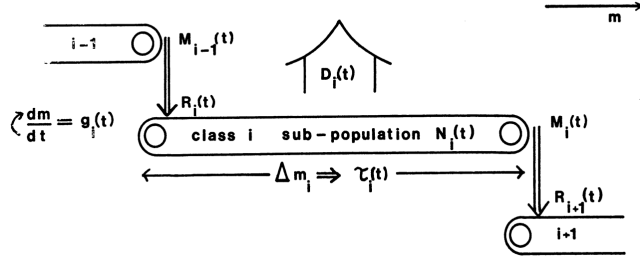


Figure 1: Caption

For this setup, consider the following equations and terms.  $R_J(t)$  is the birth rate into the juvenile stage.  $\rho_B(t)$  and  $\rho_E(t)$  are the density of individuals ( $\#/L$ ) at the beginning and end of the juvenile conveyor belt at time  $t$ .  $M(t)$  is the rate of maturation. Lastly,  $h(t)$  is the juvenile development rate, i.e how fast the conveyor belt is moving. With these terms in mind, we propose the following equalities:

$$M(t) = h(t) \cdot \rho_E(t) \quad (7a)$$

$$\rho_B(t) = \frac{R_J(t)}{h(t)} \quad (7b)$$

Equation (7a) claims that the rate of maturation, or exit from the conveyor belt, is simply the speed of the conveyor belt  $h(t)$  times the density of individuals at the end of the belt. Equation (7b) is similar, stating that the density of individuals at the beginning of the conveyor belt equals the birth rate divided by the conveyor belt's speed. Now notice that although the speed of the conveyor belt might change as the eating rate of individuals changes, an individual who matures at time  $t$  must have been born at time  $t - \tau(t)$ . Thus we can propose a third equality

$$\rho_E(t) = \rho_B(t - \tau(t)) \cdot S(t) \quad (8)$$

where  $S(t)$  is the probability of through-stage survival. Equation (8) states that the density of individuals exiting at time  $t$  must be equal to the density of surviving individuals who were born at time  $t - \tau(t)$ . Now let us plug equation (8) into equation (7a) yielding

$$M(t) = h(t) \cdot \rho_B(t - \tau(t)) \cdot S(t)$$

and then plug  $t - \tau(t)$  into equation (7b) which yields our final form

$$M(t) = R_J(t - \tau(t)) \cdot \frac{h(t)}{h(t - \tau(t))} \cdot S(t)$$

## 4 Non-Dimensionalization

$$\frac{dF(t)}{dt} = qF(t) \left(1 - \frac{F(t)}{k}\right) - \frac{F(t)}{F(t) + f_h} (I_J J(t) + I_A A(t)) \quad (9a)$$

$$\frac{dJ(t)}{dt} = \frac{\chi I_A \sigma_A}{\gamma} \frac{F(t)}{F(t) + f_h} (A(t) - A(t - \tau(t)) S(t)) - \frac{\mu_J}{I_J \sigma_J} \frac{F(t) + f_h}{F(t)} J(t) \quad (9b)$$

$$\frac{dA(t)}{dt} = \frac{\chi I_A \sigma_A}{\gamma} \frac{F(t)}{F(t) + f_h} A(t - \tau(t)) S(t) - \frac{\mu_A}{I_A \sigma_A} \frac{F(t) + f_h}{F(t)} A(t) \quad (9c)$$

$$w = \sigma_J I_J \int_{t-\tau(t)}^t \frac{F(\xi)}{F(\xi) + f_h} d\xi \quad (9d)$$

$$S(t) = \exp \left( -\frac{\mu_J}{I_J \sigma_J} \int_{t-\tau(t)}^t \frac{F(\xi) + f_h}{F(\xi)} d\xi \right) \quad (9e)$$

Equations (9) are made non-dimensional by scaling the variables as follows. Let

$$\tilde{F} = F \frac{1}{k}, \tilde{J} = J \frac{I_J}{qk}, \tilde{A} = A \frac{I_A}{qk}, \tilde{t} = tq, \tilde{\tau} = \tau q,$$

where the tilde identifies the dimensionless variable. Next, introduce the following dimensionless parameters

$$\lambda = \frac{f_h}{k}, \alpha_J = \frac{\chi}{\gamma} \frac{I_J \sigma_A}{q}, \alpha_A = \frac{\chi}{\gamma} \frac{I_A \sigma_A}{q}, \eta_J = \frac{\mu_J}{q I_J \sigma_J}, \eta_A = \frac{\mu_A}{q I_A \sigma_A}, \rho = \frac{I_J \sigma_J}{wq}$$

Now, let us make the following substitutions:

$$F = k\tilde{F}, J = \tilde{J} \frac{qk}{I_J}, A = \tilde{A} \frac{qk}{I_A}, t = \frac{1}{q}\tilde{t}, \tau = \frac{1}{q}\tilde{\tau}$$

The only substitution you need for  $t$  and  $\tau$  is scaling the differential by  $\frac{1}{q}$ . Everywhere else you just sub in those relations to get:

$$qk \frac{d\tilde{F}}{d\tilde{t}} = qk\tilde{F} \left(1 - \tilde{F}\right) - \frac{k\tilde{F}}{k\tilde{F} + f_h} \left(qk\tilde{J} + qk\tilde{A}\right) \quad (10a)$$

$$\frac{q^2 k}{I_J} \frac{d\tilde{J}}{d\tilde{t}} = \frac{\chi I_A \sigma_A}{\gamma} \frac{k\tilde{F}}{k\tilde{F} + f_h} \left( \frac{qk}{I_A} \tilde{A} - \frac{qk}{I_A} \tilde{A}(\tilde{t} - \tilde{\tau}) S \right) - \frac{\mu_J}{I_J \sigma_J} \frac{k\tilde{F} + f_h}{k\tilde{F}} \frac{qk}{I_J} \tilde{J} \quad (10b)$$

$$\frac{q^2 k}{I_A} \frac{d\tilde{A}}{d\tilde{t}} = \frac{\chi I_A \sigma_A}{\gamma} \frac{k\tilde{F}}{k\tilde{F} + f_h} \frac{qk}{I_A} \tilde{A}(\tilde{t} - \tilde{\tau}) S - \frac{\mu_A}{I_A \sigma_A} \frac{k\tilde{F} + f_h}{k\tilde{F}} \frac{qk}{I_A} \tilde{A} \quad (10c)$$

$$w = \sigma_J I_J \int_{\tilde{t}-\tilde{\tau}}^{\tilde{t}} \frac{k\tilde{F}(\tilde{\xi})}{k\tilde{F}(\tilde{\xi}) + f_h} \frac{1}{q} d\tilde{\xi} \quad (10d)$$

$$S = \exp \left( -\frac{\mu_J}{I_J \sigma_J} \int_{\tilde{t}-\tilde{\tau}}^{\tilde{t}} \frac{k\tilde{F}(\tilde{\xi}) + f_h}{k\tilde{F}(\tilde{\xi})} \frac{1}{q} d\tilde{\xi} \right) \quad (10e)$$

Now divide out  $qk$  from (10a) and divide top and bottom by  $k$  giving:

$$\frac{d\tilde{F}}{d\tilde{t}} = \tilde{F} \left(1 - \tilde{F}\right) - \frac{\tilde{F}}{\tilde{F} + \frac{f_h}{k}} \left(\tilde{J} + \tilde{A}\right)$$

Now divide by  $\frac{q^2 k}{I_J}$  from (10b) and divide top and bottom by  $k$  to get:

$$\frac{d\tilde{J}}{d\tilde{t}} = \frac{\chi I_J \sigma_A}{\gamma q} \frac{\tilde{F}}{\tilde{F} + \frac{f_h}{k}} \left(\tilde{A} - \tilde{A}(\tilde{t} - \tilde{\tau})S\right) - \frac{\mu_J}{q I_J \sigma_J} \frac{\tilde{F} + \frac{f_h}{k}}{\tilde{F}} \tilde{J}$$

Now do the same for (10c) but with  $\frac{q^2 k}{I_A}$  giving:

$$\frac{d\tilde{A}}{d\tilde{t}} = \frac{\chi I_A \sigma_A}{\gamma q} \frac{\tilde{F}}{\tilde{F} + \frac{f_h}{k}} \tilde{A}(\tilde{t} - \tilde{\tau})S - \frac{\mu_A}{q I_A \sigma_A} \frac{\tilde{F} + \frac{f_h}{k}}{\tilde{F}} \tilde{A}$$

Now for (10d) divide by  $\omega$  and combine with  $q$  and divide top and bottom by  $k$  giving:

$$1 = \frac{\sigma_J I_J}{w q} \int_{\tilde{t}-\tilde{\tau}}^{\tilde{t}} \frac{\tilde{F}(\tilde{\xi})}{\tilde{F}(\tilde{\xi}) + \frac{f_h}{k}} d\tilde{\xi}$$

Now for (10e) combine the  $q$  and divide top and bottom by  $k$  to give:

$$S(\tilde{t}) = \exp \left( -\frac{\mu_J}{q I_J \sigma_J} \int_{\tilde{t}-\tilde{\tau}}^{\tilde{t}} \frac{\tilde{F}(\tilde{\xi}) + \frac{f_h}{k}}{\tilde{F}(\tilde{\xi})} d\tilde{\xi} \right)$$

Now subbing in the dimensionless parameters and dropping the tildes, we get:

$$\frac{dF}{dt} = F(1 - F) - \frac{F}{F + \lambda} (J + A) \quad (11a)$$

$$\frac{dJ}{dt} = \alpha_J \frac{F}{F + \lambda} (A - A(t - \tau)S) - \eta_J \frac{F + \lambda}{F} J \quad (11b)$$

$$\frac{dA}{dt} = \alpha_A \frac{F}{F + \lambda} A(t - \tau)S - \eta_A \frac{F + \lambda}{F} A \quad (11c)$$

$$1 = \rho \int_{t-\tau}^t \frac{F(\xi)}{F(\xi) + \lambda} d\xi \quad (11d)$$

$$S(t) = \exp \left( -\eta_J \int_{t-\tau}^t \frac{F(\xi) + \lambda}{F(\xi)} d\xi \right) \quad (11e)$$

## 5 Substitution of Variables

Since the above system uses a variable delay in  $\tau(t)$ , we must use a change of variables to enable numerical bifurcation. To do so, let us define the new variable

$$\phi(t) = \rho \int_0^t \frac{F(\xi)}{F(\xi) + \lambda} d\xi \quad (12a)$$

$$\frac{d\phi(t)}{dt} = \rho \frac{F(\phi)}{F(\phi) + \lambda} \quad (12b)$$

$\phi(t)$  can be thought of as the amount of juvenile growth in units of individuals from time 0 to time  $t$ . And thus  $\frac{d\phi(t)}{dt}$  is the change in that growth over time.

$$\phi(t) - \phi(t - \tau(t)) = \rho \int_0^t \frac{F(\xi)}{F(\xi) + \lambda} d\xi - \rho \int_0^{t-\tau(t)} \frac{F(\xi)}{F(\xi) + \lambda} d\xi \quad (13)$$

$$= \rho \int_{t-\tau(t)}^t \frac{F(\xi)}{F(\xi) + \lambda} d\xi \quad (14)$$

$$= 1 \quad (15)$$

Thus,

$$\phi(t - \tau(t)) = \phi(t) - 1$$

which means that the birth time of a juvenile who matures at time  $t$  in units of  $\phi$ , is  $\phi(t) - 1$ .

Now, let us substitute  $\phi$  into our non-dimensional system of equations (11). Starting with (11a), let us perform that substitution, remembering to use the chain rule on the left side of the equation:

$$\frac{dF(\phi)}{d\phi(t)} \frac{d\phi(t)}{dt} = F(\phi)(1 - F(\phi)) - \frac{F(\phi)}{F(\phi) + \lambda} (J(\phi) + A(\phi)) \quad (16)$$

Now let us, rearrange (12b) to yield

$$\frac{F(\phi)}{F(\phi) + \lambda} = \frac{1}{\rho} \frac{d\phi(t)}{dt} \quad (17)$$

and plug (17) into (16) resulting in

$$\frac{dF(\phi)}{d\phi} \frac{d\phi(t)}{dt} = \frac{1}{\rho} \frac{d\phi(t)}{dt} (F(\phi) + \lambda)(1 - F(\phi)) - \frac{1}{\rho} \frac{d\phi(t)}{dt} (J(\phi) + A(\phi)). \quad (18)$$

which can be simplified to

$$\frac{dF(\phi)}{d\phi} = \frac{1}{\rho} ((F(\phi) + \lambda)(1 - F(\phi)) - J(\phi) - A(\phi)) \quad (19)$$

Now moving onto (11b), let us first substitute  $\phi$  for  $t$  recalling  $t - \tau(t)$  in units of  $\phi$  is equal to  $\phi - 1$ :

$$\frac{dJ(\phi)}{d\phi(t)} \frac{d\phi(t)}{dt} = \alpha_J \frac{F(\phi)}{F(\phi) + \lambda} (A(\phi) - A(\phi - 1)S(\phi) - \eta_J \frac{F(\phi) + \lambda}{F(\phi)} J(\phi)) \quad (20)$$



Then by again by substituting (17), we are left with

$$\frac{dJ(\phi)}{d\phi(t)} \frac{d\phi(t)}{dt} = \frac{\alpha_J}{\rho} \frac{d\phi(t)}{dt} (A(\phi) - A(\phi - 1)S(\phi)) - \eta_J \frac{F(\phi) + \lambda}{F(\phi)} J(\phi) \quad (21)$$

Next, we can divide both sides by  $\frac{d\phi(t)}{dt} = \rho \frac{F(\phi)}{F(\phi) + \lambda}$  which leaves us with the correct derivative on the left side, and squares the  $\frac{F(\phi) + \lambda}{F(\phi)}$  term while adding a  $1/\rho$  term on the right side:

$$\frac{dJ(\phi)}{d\phi(t)} = \frac{\alpha_J}{\rho} (A(\phi) - A(\phi - 1)S(\phi)) - \frac{\eta_J}{\rho} \left( \frac{F(\phi) + \lambda}{F(\phi)} \right)^2 J(\phi) \quad (22)$$

Equation (11c) can be rearranged by the same process as (11b) to yield:

$$\frac{dA(\phi)}{d\phi(t)} = \frac{\alpha_A}{\rho} A(\phi - 1)S(\phi) - \frac{\eta_A}{\rho} \left( \frac{F(\phi) + \lambda}{F(\phi)} \right)^2 A(\phi) \quad (23)$$

Lastly, let us substitute  $\phi$  into the survival probability equation (11e):

$$S(t) = \exp \left( -\eta_J \int_{t-\tau}^t \frac{F(\xi) + \lambda}{F(\xi)} d\xi \right) \quad (24)$$

To do this, recall that the delay  $\tau(t)$  in  $\phi$  units is 1 and that  $\xi$  has units of time. Now since,

$$\frac{d\phi(t)}{dt} = \rho \frac{F(\phi)}{F(\phi) + \lambda}$$

we know that

$$dt = \frac{1}{\rho} \frac{F(\phi) + \lambda}{F(\phi)} d\phi$$

and thus (11e) can be rearranged to the following

$$S(t) = \exp \left( -\frac{\eta_J}{\rho} \int_{t-\tau}^t \frac{F(\xi) + \lambda}{F(\xi)} \frac{F(\xi) + \lambda}{F(\xi)} d\phi \right) \quad (25)$$

$$= \exp \left( -\frac{\eta_J}{\rho} \int_{t-\tau}^t \left( \frac{F(\xi) + \lambda}{F(\xi)} \right)^2 d\phi \right) \quad (26)$$

Now let us compile these substitutions into a single set of equations, which is a system of fixed-delay differential equations in terms of  $\phi$ :

$$\frac{dF(\phi)}{d\phi} = \frac{1}{\rho}((F(\phi) + \lambda)(1 - F(\phi)) - J(\phi) - A(\phi)) \quad (27a)$$

$$\frac{dJ(\phi)}{d\phi(t)} = \frac{\alpha_J}{\rho}(A(\phi) - A(\phi - 1)S(\phi)) - \frac{\eta_J}{\rho} \left( \frac{F(\phi) + \lambda}{F(\phi)} \right)^2 J(\phi) \quad (27b)$$

$$\frac{dA(\phi)}{d\phi(t)} = \frac{\alpha_A}{\rho}A(\phi - 1)S(\phi) - \frac{\eta_A}{\rho} \left( \frac{F(\phi) + \lambda}{F(\phi)} \right)^2 A(\phi) \quad (27c)$$

$$S(\phi) = \exp \left( -\frac{\eta_J}{\rho} \int_{\phi-1}^{\phi} \left( \frac{F(\xi) + \lambda}{F(\xi)} \right)^2 d\xi \right) \quad (27d)$$

## 6 Survival Integral Approximation

After changing the integration variable, our survival probability is written as

$$S(\phi) = \exp \left( -\frac{\eta_J}{\rho} \int_{\phi-1}^{\phi} \left( \frac{F(\xi) + \lambda}{F(\xi)} \right)^2 d\xi \right) \quad (28)$$

Equation (28) is a state-dependent integro-differential equation. This is because each value of  $S$  depends on the integral of the function  $F(\xi)$  throughout the interval  $[\phi - 1, \phi]$ . Doing a bifurcation analysis of this requires computing a Jacobian, which is derived with respect to the function evaluated at the current and previous states. Usually, this is just 2 differential equations; however, calculating the Jacobian of equation (28) gives:

$$\frac{\partial S}{\partial F(\xi)} = S(\phi) \cdot \left( -\frac{\eta_J}{\rho} \right) \cdot \frac{\partial}{\partial F(\xi)} \left( \frac{F(\xi) + \lambda}{F(\xi)} \right)^2$$

Thus, the Jacobian is a continuous function over the range  $\xi \in [\phi - 1, \phi]$ , which is an infinite number of differential equations, which means it can't be evaluated at fixed points in a bifurcation analysis.

Instead of an infinite system of differential equations, we want to approximate the survival integral as a finite series of differential equations using low-pass filters.

We start by defining two new variables:

$$z(\phi) = \frac{1}{\zeta} \left( \frac{F(\phi) + \lambda}{F(\phi)} \right)^2$$

Isolating the integrand inside  $S(\phi)$  allows it to be expressed as:

$$\int_{\phi-1}^{\phi} \left( \frac{F(\xi) + \lambda}{F(\xi)} \right)^2 d\xi = \zeta \int_{\phi-1}^{\phi} z(\xi) d\xi$$

We now approximate  $\int_{\phi-1}^{\phi} z(\xi) d\xi$  with a chain of low pass filters defined below:

$$\frac{dQ_i(\phi)}{d\phi} = \begin{cases} -\nu Q_i(\phi) + z(\phi), & \text{if } i = 1 \\ -\nu Q_i(\phi) + Q_{i-1}(\phi), & \text{if } i > 1 \end{cases}$$

It is a low-pass filter because consider:

$$\frac{dQ_1(\phi)}{d\phi} = z(\phi) - \nu Q_1$$

If  $z(\phi) > \nu Q_1$  then  $\frac{dQ_1(\phi)}{d\phi} > 0$  meaning  $Q_1(\phi)$  increases toward  $z(\phi)$  and vice versa. So it essentially only approximates  $z(\phi)$  slowly and doesn't allow a jump from  $Q_i(\phi)$  to  $z(\phi)$ . As we consider  $i > 1$ , we get closer and closer because we have more terms fine-tuning this approximation.

Now rearrange the equation for  $\frac{dQ_1(\phi)}{d\phi}$  to get:

$$z(\phi) = \frac{dQ_1(\phi)}{d\phi} + \nu Q_1(\phi)$$

Now we integrate over a time interval which means swapping  $\phi$  to  $\xi \in [\phi - 1, \phi]$ :

$$\begin{aligned} \int_{\phi-1}^{\phi} z(\xi) d\xi &= \int_{\phi-1}^{\phi} \left( \frac{dQ_1(\xi)}{d\xi} + \nu Q_1(\xi) \right) d\xi \\ &= Q_1(\phi) - Q_1(\phi - 1) + \nu \int_{\phi-1}^{\phi} Q_1(\xi) d\xi. \end{aligned}$$

Now the same trick can be applied to  $\int_{\phi-1}^{\phi} Q_1(\xi) d\xi$  using  $\frac{dQ_2(\phi)}{d\phi}$ . This creates a series approximation of  $\int_{\phi-1}^{\phi} z(\xi) d\xi$ .

This series can be written as:

$$S(\phi) \approx \exp \left[ -\zeta \frac{\eta_I}{\rho} \left( \sum_{i=1}^n \nu^{i-1} [Q_i(\phi) - Q_i(\phi - 1)] + \nu^n Q_n(\phi) \right) \right]$$

where, after  $n$  low-pass filters, the last filter becomes negligible and can be approximated as:  $\nu^n \int_{\phi-1}^{\phi} Q_n(\xi) d\xi = \nu^n Q_n(\phi)$ .

The paper found that  $\nu = 1$ ,  $n = 5$ , and  $\zeta = 1000$  were sufficient to approximate.

Now the survival integral is expressed only in terms of  $(\phi)$  and  $(\phi - 1)$ , which means the Jacobian is finite and a bifurcation analysis can be performed.

## 7 Reproduction of Nature (2008) Supplement Results

In the supplemental information to the Nature (2008) paper, after making a substitution of variables as explained in section 5 and using the survival integral approximation as explained in section 6, McCauley produces Figure 2.

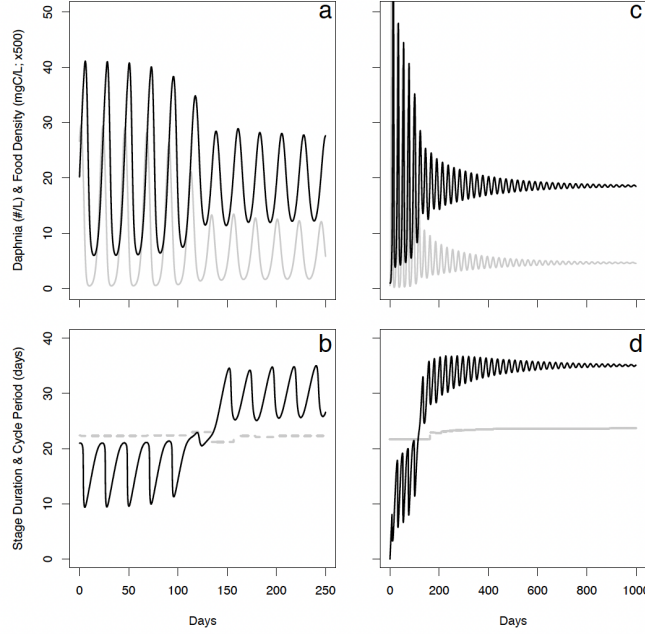


Figure 2: Transient consumer-resource dynamics. a)-b) The transition from large-amplitude to small-amplitude cycles was always found to be abrupt. For these simulations,  $K = 0.6$  and initial conditions were chosen close to the unstable limit cycle. c)-d). Since the steady state is stable outside the region of limit cycles, we demonstrate the influence of all attractors at  $K = 0.3$  and with initial conditions far from equilibrium. Even though large-amplitude and small-amplitude attractors do not exist in this part of parameter space, the consumer-resource dynamics remain influenced by their basins of attraction. Dynamics begin with damped large-amplitude cycles followed by an abrupt switch to damped small-amplitude cycles. The remaining transient dynamics to steady-state are characterized by damped oscillations. Lines have the same meaning as Figure 1.

We wanted to reproduce these figures, so we coded our own versions using equations (27) and the survival-probability approximations in 6. The supplement did not provide full history functions for these figures, so we estimated constant histories from the figures themselves<sup>1</sup>. We use the same parameters from the Nature paper (see Table 3). We have two versions of the code, one from Milan and one from Arno, available at [GitHub](#). Here is our recreated figure from Arno's code: `nature_supplement_recreation.m`:

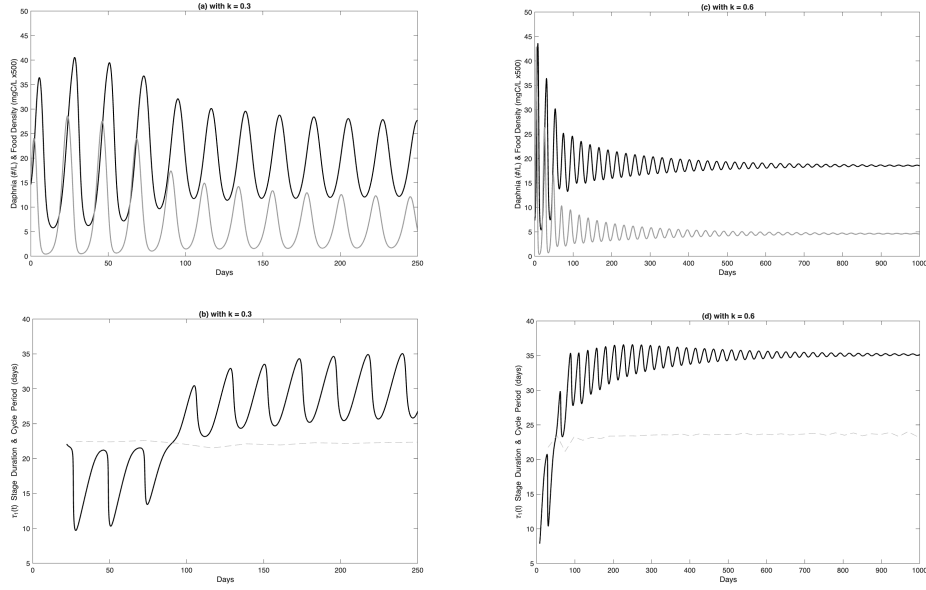


Figure 3: Recreation of Figure 2. Daphnia population ( $\#/L$ ) in black, Algae Density ( $\text{mgC}/L \times 500$ ) in gray, cycle period (days) with a dashed line and stage duration (days) with a solid line

Figure 3 match relatively well to Figure 2 as they show the same general shape and behavior. Without specifically defined history functions, we are unable to match much better.

<sup>1</sup>These histories can be seen in the history vector within our code

## 8 Algebraic Equilibrium Setup for Code

We start by finding  $F^*$ . At equilibrium, all of the differential equations are equal to zero:

$$\frac{dF^*}{d\phi} = \frac{dJ^*}{d\phi} = \frac{dA^*}{d\phi} = \frac{dQ_i^*}{d\phi} = 0$$

We start with the adult equation at equilibrium with the goal of getting an expression that we can solve in terms of  $F^*$ . The delay has no effect so we have:  $A(\phi - 1) = A(\phi) = A^*$

Thus, the adult equation becomes:

$$0 = \frac{\alpha_A}{\rho} A^* S(\phi) - \frac{\eta_A}{\rho} \left( \frac{F^* + \lambda}{F^*} \right)^2 A^*$$

Since we are looking for positive equilibrium we assume  $A^* > 0$  so we can divide out  $\frac{A^*}{\rho}$  giving:

$$0 = \alpha_A S(\phi) - \eta_A \left( \frac{F^* + \lambda}{F^*} \right)^2 \quad (29)$$

Additionally,  $S(\phi)$  changes at equilibrium, all  $Q_i^*$  are the same by solving  $0 = \frac{dQ_i^*}{d\phi} = \frac{dQ_{i-1}^*}{d\phi}$ :

$$S(\phi) \approx \exp \left[ -\zeta \frac{\eta_J}{\rho} \left( \sum_{i=1}^n \nu^{i-1} [Q^* - Q^*] + \nu^n Q^* \right) \right] = \exp \left[ -\zeta \frac{\eta_J}{\rho} \nu^5 Q^* \right]$$

We know that  $0 = \frac{dQ_1^*}{d\phi} = \frac{dQ^*}{d\phi} = -\nu Q^* + \frac{1}{\zeta} \left( \frac{F^* + \lambda}{F^*} \right)^2$

Which can be rearranged to give:

$$Q^* = \frac{1}{\zeta \nu} \left( \frac{F^* + \lambda}{F^*} \right)^2$$

Now since  $\nu = 1$  we can plug in to get  $S$  now as a function of  $F^*$ :

$$S(F^*) \approx \exp \left[ -\frac{\eta_J}{\rho} \left( \frac{F^* + \lambda}{F^*} \right)^2 \right] \quad (30)$$

Thus we can now rewrite equation 29 only in terms of  $F^*$  below:

$$0 = \alpha_A \exp \left[ -\frac{\eta_J}{\rho} \left( \frac{F^* + \lambda}{F^*} \right)^2 \right] - \eta_A \left( \frac{F^* + \lambda}{F^*} \right)^2 \quad (31)$$

Since  $\lambda = \frac{f_h}{K}$  we can solve this to find  $F^*$  for a given value of  $K$ .

Next, we find  $A^*$  by starting with algae equation at equilibrium with the idea of getting an equation for  $A^*$ . This gives:

$$0 = \frac{1}{\rho}((1 - F^*)(F^* + \lambda) - J^* - A^*)$$

Which simplifies to:

$$A^* = (1 - F^*)(F^* + \lambda) - J^* \quad (32)$$

Now we solve for the juvenile equation at equilibrium to get an expression for  $J^*$  to plug into equation 32:

$$0 = \frac{\alpha_J}{\rho}(A^* - A^*S^*) - \frac{\eta_J}{\rho} \left( \frac{F^* + \lambda}{F^*} \right)^2 J^*$$

Which reduces to:

$$J^* = \frac{\alpha_J}{\eta_J} \left( \frac{F^*}{F^* + \lambda} \right)^2 (1 - S^*)A^* \quad (33)$$

$$A^* = (1 - F^*)(F^* + \lambda) - \frac{\alpha_J}{\eta_J} \left( \frac{F^*}{F^* + \lambda} \right)^2 (1 - S^*)A^*$$

Thus, we can combine terms and simplify into an equation for  $A^*$  in terms of just  $F^*$  which is already found in equation 31:

$$A^* = \frac{(1 - F^*)(F^* + \lambda)}{1 + \frac{\alpha_J}{\eta_J} \left( \frac{F^*}{F^* + \lambda} \right)^2 (1 - S^*)} \quad (34)$$

where  $S^*$  is equation 30 which is in terms of just  $F^*$ .

Finally, now that we know  $A^*$ , we find  $J^*$  just from equation 33.

Since  $\lambda$  changes with  $K$ , these equilibrium equations can be plotted against  $K$  to see how the analytic equilibrium changes with respect to  $K$ .

## 9 Equilibrium Behavior

### 9.1 Linear Stability

To investigate stability, it is standard to perform a linear stability analysis. For this, we make infinitesimal perturbations around the linearized system. We implemented MATLAB's BIFTOOL software, allowing us to find the steady-state branch of equilibrium solutions using an extrapolation and Newton correction method. Then a BIFTOOL function allows us to linearize each equilibrium solution. It does this by setting up a matrix representing the response to an infinitesimal perturbation, then calculates the eigenvalues. These are the characteristic roots of each solution. To evaluate whether an equilibrium solution is linearly stable or unstable, we use the fact that an equilibrium point is stable if and only if the real part of all of its characteristic roots is negative. Any positive real parts result in an unstable equilibrium point. Thus, we can simply

extract the maximum real part of characteristic roots for each equilibrium solution. It is important to note that solutions to delay differential equations are infinite-dimensional, resulting in transcendental characteristic equations. This means that, in theory, there are an infinite number of characteristic roots for each solution; however, BIFTOOL simply calculates a finite and controllable subset of these characteristic roots.

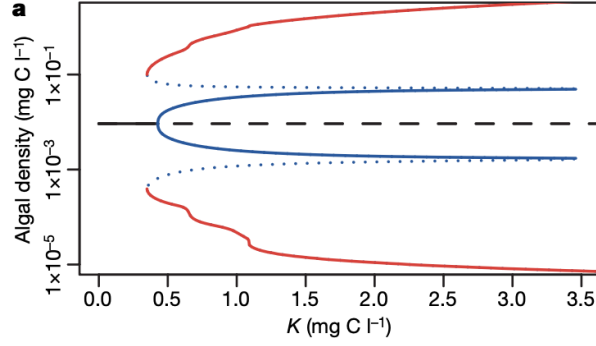


Figure 4: Figure 1a from Nature (2008)

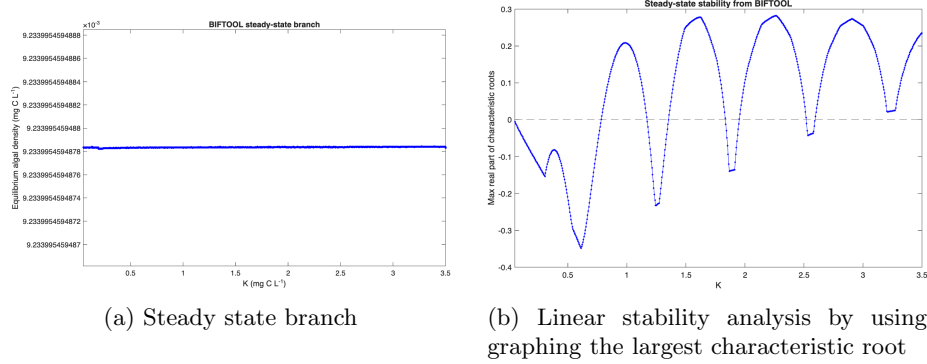


Figure 5: Steady-state solutions and stability showing hopf bifurcations

Figure 8b displays alternating linear stable and then unstable behavior. This is done by graphing the largest real part of the characteristic roots against the  $K$  value for the equilibrium solution. Thus, we can confidently say that an equilibrium solution is unstable if that maximum value is negative and vice versa if it is positive. When comparing this with Figure 1a in the Nature (2008) paper, we see similar behavior in the steady-state values, indicating that the BIFTOOL branch continuation method was accurate but not in the linear stability analysis. In Figure 1a, they characterize the steady-state solutions for  $K$  values from 0-0.5 as stable, then after undergoing a Hopf bifurcation at  $K$



$= 0.5$ , all solutions become unstable. However, our results show a series of Hopf bifurcations indicating that there should be sections of stability after that initial bifurcation. Points where the max real part of the characteristic root passes from positive to negative are Hopf bifurcations, as these points signify when a complex conjugate pair of roots cross the imaginary axis. This trend continued even after expanding the window of characteristic root calculation and other debugging of the code, which suggests an issue in this implementation or model.

## 9.2 Perturbation Stability

To investigate equilibrium behavior, we perform a perturbation stability analysis on the equilibrium solutions to see if they can be classified computationally as stable or unstable at each value of  $K$ .

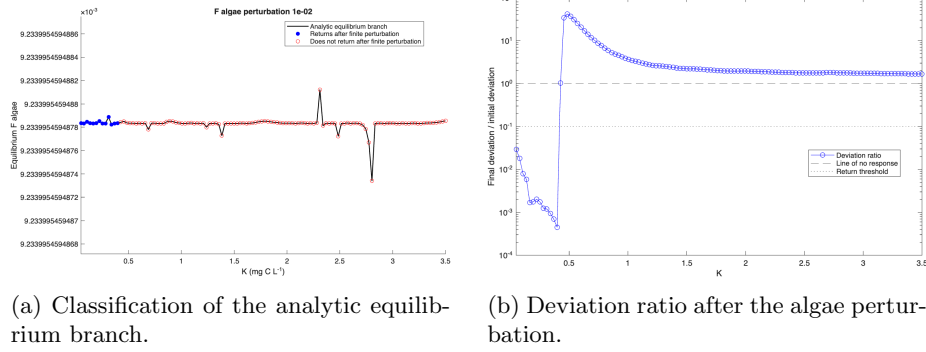


Figure 6: Response to a 1% algae perturbation. Points are defined as returning when the late-time deviation from equilibrium is less than 10% of the early-time deviation. The calculated first  $K$  value that did not return is: 0.426891

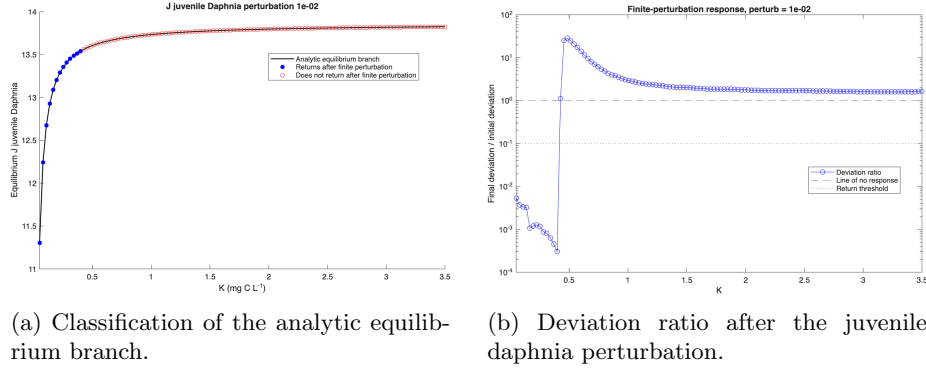


Figure 7: Response to a 1% adult daphnia perturbation. Points are defined as returning when the late-time deviation from equilibrium is less than 10% of the early-time deviation. The calculated first  $K$  value that did not return is: 0.426891

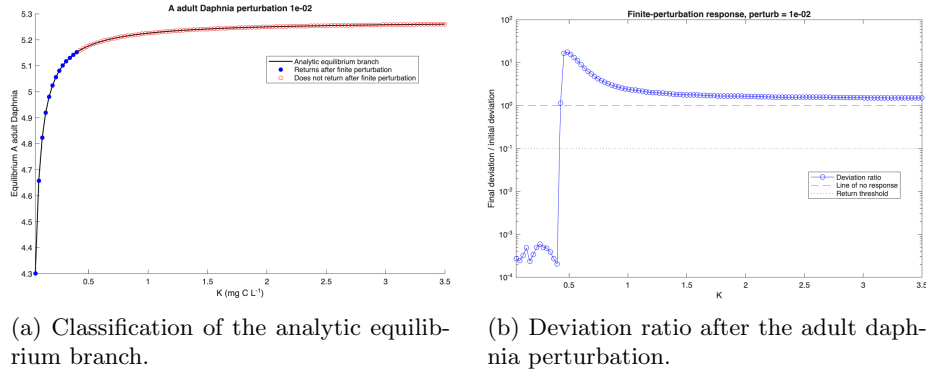


Figure 8: Response to a 1% adult daphnia perturbation. Points are defined as returning when the late-time deviation from equilibrium is less than 10% of the early-time deviation. The calculated first  $K$  value that did not return is: 0.426891

Classifications of stability are found from calculating a perturbation response via a deviation ratio, meaning it is not linear stability. However, we can characterize equilibrium points as stable for some finite perturbation, in this case we had a 1% perturbation of each population and an arbitrary threshold of 10% to characterize a return response. The pattern of finite-perturbation stability for  $K$  values from 0-0.5 followed by a bifurcation to instability matches those from figure 4 which depicts the Nature (2008) bifurcation plot along with cycle behavior. Thus, the perturbation stability analysis seems to better match the steady-state behavior seen in the Nature (2008) paper.

In Figures 7, 8, and 9, we can see a balancing effect where an increase

in carrying capacity in the algae doesn't affect the equilibrium density value. Instead, the equilibrium of algae remains constant at about 0.01 (mgC/L) for all values of  $K$ . In response, we see a logarithmic growth in the equilibrium solutions of both Daphnia populations. We interpret this as an increase in carrying capacity that allows the algae density to increase; however, the Daphnia increase consumption in response, keeping the algae density constant. Thus, the equilibrium for algae remains unchanged, while both Daphnia equilibrium values increase logarithmically with carrying capacity. The logarithmic shape may indicate the carrying capacity hitting an effective limit where increasing the carrying capacity no longer contributes to algae density due to other limiting factors.

The next step is trying to recreate the small- and large-amplitude cycles pictured in blue and red in Figure 7. To do this, we use a BIFTOOL function that creates a small oscillation around a Hopf bifurcation. While it doesn't line up with  $K=0.5$  as seen in Figure 7, the characteristic root method showed the Hopf bifurcation to be around  $K \approx 0.78$ . Thus, we use the function to create the cycle around that first Hopf bifurcation to see if it shows similar dynamics to the first bifurcation in Figure 7. Once the oscillation is made, the continuation process proceeds by guessing the next cycle based on the past cycles, then Newton correcting it so it satisfies the DDE model, then recording the amplitude and other data. The correction looks like making sure each cycle derivative matches the original DDE model for that  $K$  value.

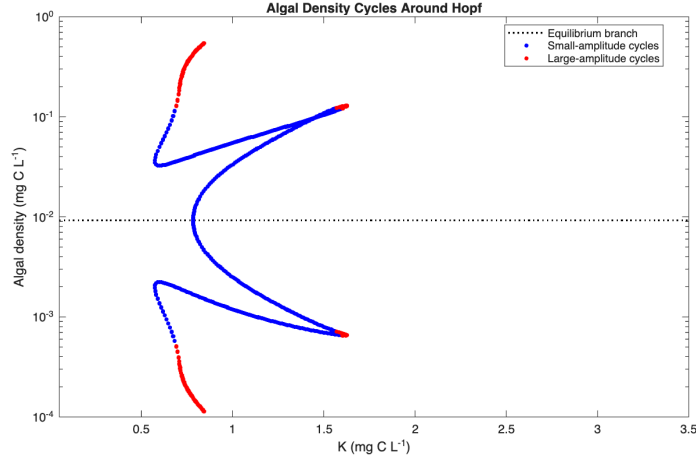


Figure 9: Branch from Hopf Bifurcation at  $K \approx 0.78$  showing cycle behavior

Figure 9 displays the result of the branch starting from the first Hopf bifurcation at  $K \approx 0.78$ . In order to crudely replicate the division of small- and large-amplitude cycles, an arbitrary amplitude threshold of 0.12 was chosen, where cycles with smaller amplitudes were labeled as small-amplitude cycles in blue and cycles with larger amplitudes were large amplitude cycles in red. The resulting figure shows an odd shape but one resembling the multi-tiered struc-

ture of Figure 7. The reason the branch stops around  $K \approx 1.7$  then curves back for the large amplitude cycle is that the large and small amplitude cycles essentially become the same cycle at that fold. Since our dubious linear stability analysis showed that there are other Hopf bifurcations, this branch only accounts for the cycles around the Hopf values  $K \approx 0.78$ , which extend to the fold. Thus, this branch only accounts for one section of the full dynamics, and plotting with the cycles around the other Hopf bifurcations may be a way to get the full diagram. However, even with debugging and other attempts to get the full cycle behavior, this was the best result, indicating that this method is not what McCauley et. al. used to get their figure. Additionally, the method of thresholding into small- and large-amplitude cycles should be better-defined. In the Nature (2008) paper the small- and large-amplitude cycles are distinguished by “the ratio of the cycle period to juvenile stage duration” instead of simply choosing an arbitrary value to threshold density values in the plot. The similarity in the general shape is worth noting despite the inconsistency.

In the future, the consistency of these stability analyses with Nature (2008) needs to be improved. These results need further refinement but the ideas and implementation will hopefully lay a groundwork for future investigation.

## A Parameter Table for Figure 3

| Parameters | Value                 | Description                                                                  | Source |
|------------|-----------------------|------------------------------------------------------------------------------|--------|
| $q$        | 1.0                   | Maximum per-capita rate of resource growth ( $\text{day}^{-1}$ )             | 1      |
| $k$        | 0–5.0                 | Resource carrying capacity (C)                                               | –      |
| $I_J$      | $5.23 \times 10^{-3}$ | Maximum juvenile ingestion rate* ( $\text{C N}^{-1} \text{d}^{-1}$ )         | 2      |
| $I_A$      | $1.91 \times 10^{-2}$ | Maximum adult ingestion rate* ( $\text{C N}^{-1} \text{d}^{-1}$ )            | 2      |
| $f_h$      | 0.164                 | Half-saturation constant in functional response (C)                          | 3      |
| $\chi$     | 0.77                  | Proportion of utilized carbon allocated to reproduction (–)                  | 2      |
| $\gamma$   | $1.51 \times 10^{-3}$ | Carbon required to produce one new offspring ( $\text{C N}^{-1}$ )           | 2      |
| $\sigma_J$ | 0.49                  | Proportion of ingested carbon that is utilized by juveniles <sup>†</sup> (–) | 2      |
| $\sigma_A$ | 0.43                  | Proportion of ingested carbon that is utilized by adults <sup>†</sup> (–)    | 2      |
| $\mu_J$    | $1.09 \times 10^{-5}$ | Juvenile mortality scalar <sup>‡</sup> ( $\text{C N}^{-1} \text{d}^{-2}$ )   | 3      |
| $\mu_A$    | $5.92 \times 10^{-6}$ | Adult mortality scalar <sup>‡</sup> ( $\text{C N}^{-1} \text{d}^{-2}$ )      | 3      |
| $w$        | $4.8 \times 10^{-3}$  | Mass gain required to complete juvenile development ( $\text{C N}^{-1}$ )    | 2      |

where the dimension symbol N denotes  $\#/\text{L}$ , C denotes  $\text{mgC}/\text{L}$ , and d denotes day.

Table 3: Parameter estimates based on *Daphnia pulex* and *Chlamydomonas reinhardtii* at 24 °C